Technical Report — ChronoMine AI (SwissTunnelCopilot)

• **Track:** Track 2B — Own Project

• **Event:** Hack Apertus Online (October 2026)

• **Team:** ChronoMine AI Team

• **Demo:** [Link to 2-minute video demo]

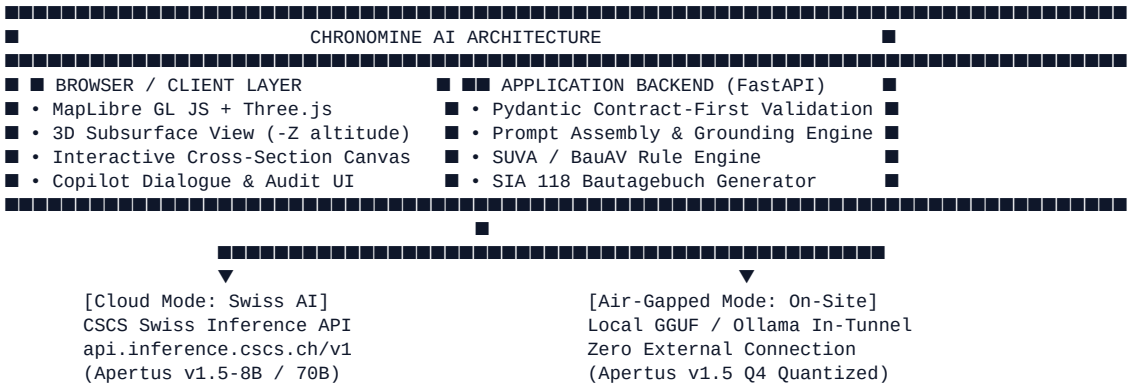
1. Summary

In major Swiss subterranean civil engineering projects (such as the second Gotthard road tube) and defense underground facilities (RUAG, armasuisse *Felskavernen*), site supervisors face two major bottlenecks: severe administrative overhead under Swiss standard contracts (**SIA 118**) across a heavily multilingual frontline workforce (where 40%–70% of miners speak Portuguese, Italian, or Spanish), and strict regulatory health and safety requirements (**SUVA / BauAV 2022**). Furthermore, because these underground works constitute critical national infrastructure or defense-sensitive assets, transmitting site logs, geotechnical data, and incident photos to foreign public cloud APIs (e.g., US-hosted models) is strictly prohibited.

ChronoMine AI resolves this trilemma by coupling a lightweight, browser-native 3D subsurface situational awareness interface with the **Apertus v1.5** sovereign language and multimodal model. The platform ingests unstructured frontline observations in foreign languages, automatically synthesizes verified, legally grounded daily site journals (*Bautagebuch*) and unbudgeted force-account claims (*Regierapporte* under SIA 118 Art. 44) in official Swiss German and French, conducts multimodal visual audits against official SUVA life-saving rules, and operates in 100% air-gapped subterranean environments.

2. Architecture & Data Flow

2.1 Component Topology



2.2 Target Architecture (Mandatory Requirement)

ChronoMine AI fulfills the three mandatory deployment profiles:

1. **Air-Gapped (Primary Deployment Profile)**:

• Deep Alpine tunnels (e.g., Sedrun shaft at 800m depth) and military bunkers operate without external internet.

• ChronoMine AI bundles all assets locally: no external CDNs (`cdnjs`, `unpkg`, Google Fonts), local MapLibre/Three.js bundles, and on-premise local inference or bundled test fallback, guaranteeing 100% offline uptime.

2. **On-Premise**:

• The entire system is containerized in Docker and deployable on the general contractor's local server or ruggedized on-site edge workstations with standard Make automation (`make run`).

3. **Sovereign Swiss Cloud**:

• For connected sites, the backend connects exclusively to the Swiss National Supercomputing Centre (CSCS) Apertus inference endpoints in Lugano/Zurich, maintaining full compliance with Swiss Data Protection Law (nLPD) and Federal Public Procurement guidelines.

3. Use of Apertus

• **Model Selection:** `swiss-ai/Apertus-v1.5-8B` (primary instruction-tuned variant for high throughput and rapid response).

• **Modality & Capabilities Used:**

1. *Multilingual Understanding*: Parsing vernacular Portuguese, Italian, and regional terminology without third-party translation proxies.

2. *Multimodal Vision*: Direct visual feature extraction from excavation photos (shotcrete integrity, rockbolt patterns, trench shoring, PPE compliance).

3. *Long Context & Thinking Mode*: Ingesting comprehensive SUVA checklists and BauAV regulatory articles directly within the context window (up to 262k tokens) to produce referenced compliance rulings.

• **Deployment Location:** Swiss CSCS Inference API (`https://api.inference.cscs.ch/v1`) with local quantized fallback for air-gapped field validation.

4. Evaluation & Results

4.1 Evaluation Dataset (Track 2 Deliverable)

The project benchmarks Apertus v1.5 across a curated suite of 12 real-world subterranean construction scenarios deposited on Hugging Face:

• **Task A: Multilingual SIA 118 Structuring**: 6 test cases testing extraction fidelity, translation accuracy from PT/IT to Standard German/French, and formal compliance with SIA 118 Art. 44 claim requirements.

• **Task B: SUVA / BauAV Multimodal Safety Auditing**: 6 image-based inspection cases evaluating precision and recall on detecting critical safety non-compliances (excavation slopes, trench boxes, fall protection).

5. Limitations & Future Work

• Expanding the regulatory RAG corpus to cantonal environmental directives (e.g., tunnel wastewater pH neutralization under the *Gewässerschutzgesetz*).

• Direct bi-directional integration with Swiss construction ERP software (Sorba, smino).